

CIS 522 – Final Project – Technical Report

Team: Am I The Overfitter

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Team Members:

- Ko-Yen Huang; kyhuang; Email: `kyhuang@seas.upenn.edu`
- Josh Ludan; jludan; Email: `jludan@seas.upenn.edu`
- Matthew Pressimone; mpressi; Email: `mpressi@seas.upenn.edu`

Abstract

Recent advances in natural language processing have shown the ability of models to accomplish increasingly complex tasks such as solving high level competition math problems and writing code at the level of an average programmer. Alongside this, recent concerns about these models have arisen due to the potential of these models to generate malicious or harmful content. To combat this, the concept of value alignment has been proposed as a way to ensure that these models only pursue goals which are beneficial to humans. In this paper, we investigate the efficacy of utilizing deep neural networks for moralistic judgment predictions with the “Am I the Asshole” Reddit forum as our training data. We aim to develop models that make moral judgments that align with the Reddit user community. Our results demonstrate the potential of using posts in social media communities like Reddit AITA forum as a dataset to benchmark value alignment in intelligent agents with existing NLP techniques. We evaluate 6 different models and find that a fine-tuned Roberta model achieves the best performance with an AUC score of 0.7 and an accuracy of 74% which is 1% above random baseline. The paper demonstrates at the same time the difficulty of the labeling task and the potential this dataset has as a future benchmark to evaluate different language models in their capability to replicate the collective decision made by a given community.

1 Introduction

Value alignment is an important property of intelligent agents. This property indicates that an intelligent agent can only pursue goals and activities which are beneficial to humans [1]. Russell [2], Moor [3], and others have argued that value alignment is one of the most important tasks facing AI researchers today.

Ideally, a value-aligned algorithm should make decisions that align with human decisions in similar situations and, in theory, make decisions which are unlikely to be harmful [4].

However, value alignment is a rather complicated task. First, it is very hard to directly specify values for the machine given a specific circumstance because there are infinite undesirable outcomes in a real open world [5]. Also, subtle modifications of a seemingly trivial aspect of a script, such as changing the gender of the subject, can change the context of the story greatly, further leading to entirely different desirable human decisions and values. This demonstrated the sophistication of making decisions or evaluating values in real world circumstances, and solutions are needed to train machines to make decisions or evaluate values like humans.

One approach to value alignment is making the machine learn through imitation learning [6]. However, achieving value alignment via imitation learning can be challenging, because learning knowledge from demonstrations that generalizes beyond the context of the observation is difficult, and it can be time-consuming to provide sufficient demonstrations. Furthermore, it can be difficult for humans to provide high quality demonstrations that exemplify certain values, especially those related to negation or not doing something [7].

An alternative approach to achieve value alignment is having the intelligent agents learn from naturally occurring stories which encode societal norms. A previous study by Sultan used children’s educational comic strip, Goofus Galant, as training data to train models to infer human values and social norms [7]. Researchers at the Allen Institute of AI at the University of Washington trained Delphi, a model that also outputs a moral judgment given text inputs, on texts in Commonsense Norm Bank [8]. Moral judgment is dependent on cultural contexts, and different communities are likely making moral judgments differently. Thus, one caveat of training a model on text datasets labeled with moral judgements by diverse communities like Commonsense Norm Bank is that the values learned by the model may not align to the values of the community that we care about most.

To train a language model that makes moral judgments that align to a more specific community, we attempt to achieve value alignment by training our language model on posts on the “Am I the Asshole” (AITA) forum. AITA is a Reddit forum where users describe a moralistic scenario that they were involved in. Other users then make a moralistic judgment about whether the original poster is an “asshole” based on the text provided. By training our language model on AITA posts, we aim to develop models that make moral judgements that align with the Reddit user community.

In summary, our work investigates the efficacy of utilizing deep neural networks for moralistic judgment predictions. The task is to predict the moralistic

judgment (Asshole or Non-asshole) made by the majority of other users given the original posts. We used LGBM as our baseline model, then compared and analyzed the efficacy of five deep learning models including fine-tuned BERT, GPT-3, SBERT, convolutional neural network, and Bi-LSTM. Based on our results, a fine-tuned Roberta model achieved the best AUC of 0.70. Our study demonstrates the potential of using posts in social media communities like Reddit AITA forum as a source of training data for achieving value alignment in intelligent agents with existing NLP techniques.

In recent years, large neural networks trained for language understanding and generation have achieved impressive results across a wide range of tasks. In 2021, Google Research introduced the Pathways Language Model (PaLM). This 540-billion parameter Transformer model was evaluated on hundreds of language understanding and generation tasks, and achieved state-of-the-art few-shot performance across most tasks [10]. DeepMind created AlphaCode that writes computer programs at a competitive level. By utilizing transformer-based language models, AlphaCode achieved an estimated rank within the top 54% of participants in programming competitions [11]. Microsoft Research also built a neural theorem prover Lean that learned to solve a variety of challenging high-school Olympiad problems [12]. Despite breakthroughs of NLP applications on various tasks, NLP applications on value alignment tasks still leave a lot to be improved. Thus one additional motivation of our project is to evaluate how different language models perform on this difficult task. We expect different models to capture different information in the same input texts during their learning process.

2 Related Work

Previous works have demonstrated the use of naturally occurring stories as training data for value alignment tasks. A previous study by Sultan used children’s educational comic strip, Goofus & Gallant (G & G), as training data to train models to infer human values and social norms [7]. Their models learned to classify sentences from G & G stories as normative or non-normative with a high degree of accuracy. Their research also provided evidence that transformer-based models trained on such naturally occurring story corpus are highly accurate in classifying previously unseen descriptions of normative behavior taken from that comic strip.

Researchers at the Allen Institute of AI trained Delphi, a model that makes moral judgements given text inputs, on texts in CommonSense Norm Bank [8]. Delphi demonstrated strong promise of language-based commonsense moral reasoning, with up to 92.1% accuracy vetted by humans, which significantly outperformed the zero-shot performance of GPT-3 of 52.3%. This suggests that massive scale alone does not endow pre-trained neural language models with human values, and the selection of data used for training plays a critical role

for value alignment. However, training models on text datasets labeled with moral judgments by diverse communities may over-generalize moral judgements across different cultural contexts, ignoring the fact that different communities are likely making moral judgments differently.

In contrast to Delphi, our team focused on training a language model that makes moral judgments based on the posts or texts of a specific community that we care about most. We specifically trained our language model on posts in the “Am I The Asshole” Reddit forum, which may better represent the Reddit community that we want to focus on. Delphi also provides an interface that would allow us to also use it as a benchmark to compare our model against to see if our specific model would be able to beat their general model which was presumably trained with more resources than those we have access to.

A previous project by Ivy Wang at Stanford University tackles the same problem we’re addressing by training DNNs using word2vec on the posts in the “Am I The Asshole” forum as the training data [9]. The models trained in this project include a “Post2Vec with SVM” model and a “Post2Vec with DNN”, but did not utilize the large pre-trained language models like BERT or SBERT. In this study, the potential of P2V-DNN remains unrealized because of a high variance of the model due to insufficient amount of training data. Since getting more training data from AITA forum may be hard to come by due to the limitations of the Reddit API, fine-tuning large pre-trained language models to adapt to this same task may be a more feasible approach. In our project, we utilized large pre-trained models like BERT, SBERT, and GPT-3 to tackle the challenges encountered by Ivy Wang. The F1 score achieved by Ivy Wang (0.31) would be a benchmark to evaluate the performance of our model. Given that our model will be leveraging a pretrained model, we hypothesize that our model will outperform hers.

3 Dataset and Features

Our dataset is a CSV file containing posts and their relevant information from Reddit’s AITA forum. The features include:

Feature Name	Description
title	The post title.
body	The main text of the post.
edited	A timestamp of when the post was edited, or False if it was not edited.
num_comments	The number of comments on the post.

The verdict column indicates the other users’ decision on who was at fault, based on four choices:

Verdict Type	Did the writer act unethically?	Did another party act unethically?
Asshole	Yes	No
Not the asshole	No	Yes
Everyone sucks	Yes	Yes
No assholes here	No	No

Our label, `is_asshole`, is 1 when the verdict is “asshole” or “everyone sucks”, and is 0 when the verdict is “not the asshole” or “no assholes here”.

For our exploratory data analysis, we started by checking for class imbalance. We found that 71117 (27.2%) posts had label 1, while 26511 (72.8%) posts had label 0, indicating a significant class imbalance.

Next, we looked at the word count of the posts (see figure 1).

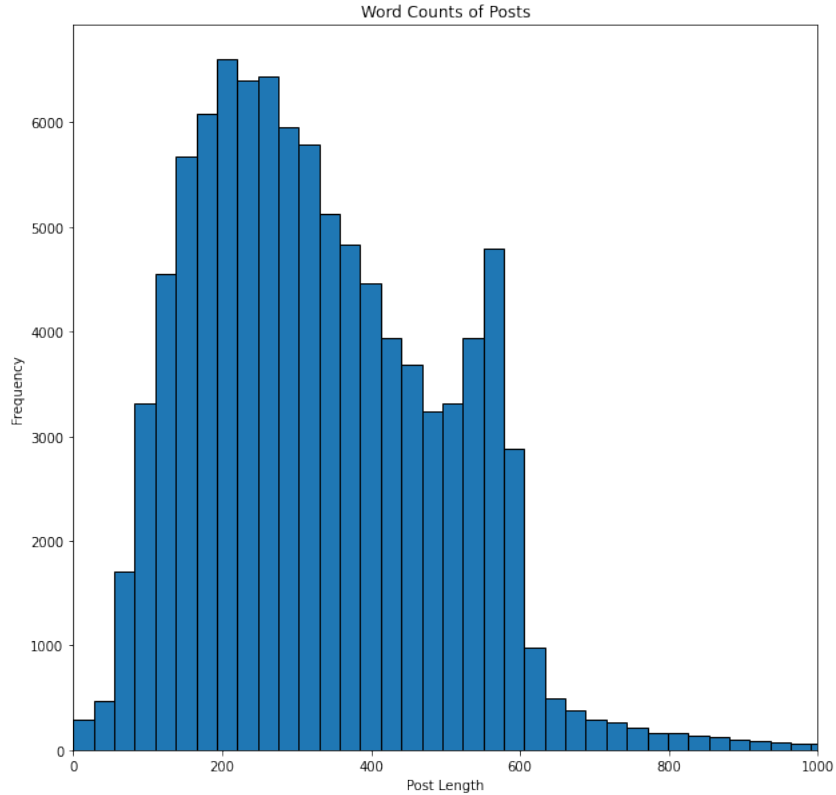


Figure 1: Word Counts of Posts

We found that posts of around 200 to 300 words were most common, posts a little under 600 words were fairly common, and posts well above 600 words were very uncommon. From this information, we can see that some of our models

that have token limits might lose a great deal of the information in some of the posts, potentially including the reasonably common 550-600 word posts. For example, consider a long post that got a label of 1 only because of one small detail at the very end. Such a post will be far easier to predict incorrectly on models with token limits, since these models will miss that tiny detail entirely and see the remaining post (that would look almost certainly like a label 0 post). In general, for posts where one or two sentences make the label what it is, token limits increase our chance of missing out on that part of the post.

We then checked the correlation between the label and numeric features using the correlation heatmap in figure 2. Note that for this task, we included the post length feature and added another feature, `is_edited`, that is 1 if the post was edited and 0 otherwise. From the heatmap, we see no correlation between anything except for an understandably high correlation between the number of comments and the score. If more people voted on the post, it means that a lot of people saw the post, indicating that the post had more people who could have commented too. Also noteworthy is the slight correlation between a post being edited and the label being 1; it makes sense why there should be little correlation (anyone could edit their post to add more information they forgot about), but notably posts that initially get voted “YTA” might have more incentive to edit their posts in order to try and change commenters’ minds. Since the edits can be made after the label is decided, this could make classification harder as, for example, a post labeled “YTA” could actually have relevant info that, if it was present before editing, could have made it an “NTA” post.

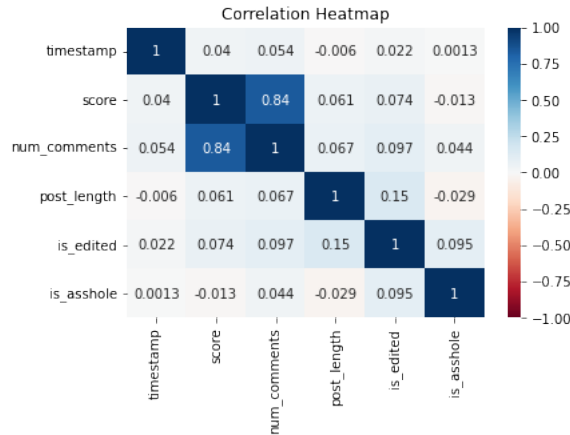


Figure 2: Correlation Heatmap

We also were interested in knowing whether or not seeing certain words in a post would be indicative of the label. To do this, we constructed a document-term matrix, fit a logistic regression using the words as features, and examined the coefficients. The most influential words and their corresponding coefficients are shown in figures 3 and 4.

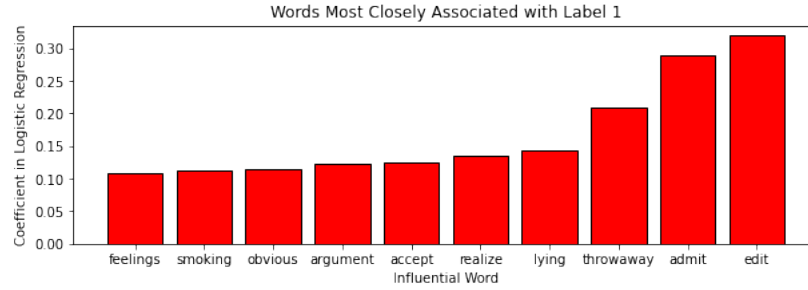


Figure 3: Most Influential Words (Label 1)

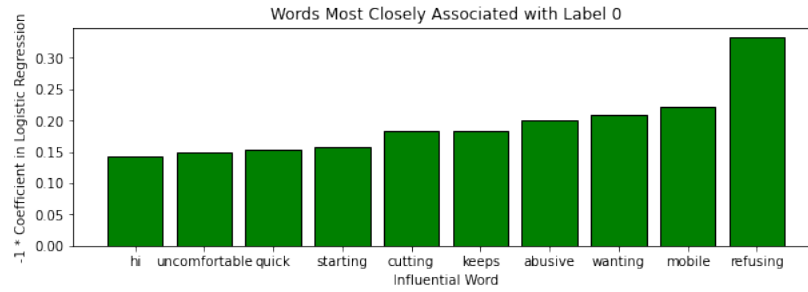


Figure 4: Most Influential Words (Label 0)

There are a few potential reasons why some of these words are seen as more indicative of the label. For example, the word “refusing”, the most influential word for label 0, could have been used when someone refuses to stop doing something that makes the post writer “uncomfortable” (another influential word for label 0). Words like “hi” and “mobile” seem out of place, but people posting from their phones often apologize for the formatting of their post because they are “on mobile.” So these words show interest in etiquette and in the people reading the post, and could then suggest that this person is less likely to act unethically.

The words that were associated with label 1 were more interesting. It is a bit surprising to see “edit” at the top of this list because the heatmap showed a negligibly small correlation between the label and whether or not the post was edited. Yet this correlation was slightly positive (i.e. there was a slight correlation between the post being edited and having label 1). Also recall that, because some posters who got “YTA” might believe they did nothing wrong, they have more incentive to add to their post (which is often seen at the end of the post, starting with “edit:”). Less surprising are “admit”, “realize,” and “accept”; if these are used in a context in which the poster admits that there was something they did that they did not feel good about, it is possible that

this action is what gave the post label 1.

Note that the coefficients associated with these words are small, but the effects are fairly significant. Using a word like “edit” once would multiply the odds of the post having label 1 by a factor of about $e^{0.3}$, or around 1.35.

From the EDA, we observe that simple models can gain some amount of predictive power by being able to pick up on shallow linguistic signals such as the presence of the word “refusing” or “edited”. A more sophisticated model may be able to perform better by being able to parse context. For example, a post in which the writer keeps “refusing” to do something unethical should have label 0, but a post in which the writer keeps “refusing” to do something that they need to do should have label 1. The ability of recent models to identify longer-range text dependencies and to model their impact on the label generated makes it interesting to ask how well they can perform better in comparison to less sophisticated learners which do not take into account relationships between tokens.

4 Methodology

4.1 Non-DL Benchmark: Gradient Tree Boosting

We tried both XGBoost and LightGBM as non-deep learning benchmarks, without any hyperparameter tuning. As both had identical results and LGBM ran faster, we only considered the latter.

4.2 Convolutional Neural Network

Here, we used a series of 1-D convolutions that output the binary prediction at the end. We found that increasing the number of convolutional layers from 3 to 7 helped, but only slightly.

4.3 LSTM

This method involved an LSTM (or BiLSTM) layer on the embedded post title and body. This was followed by some linear layers (with ReLUs between each one). As it was not properly training at first, there was a great deal of hyperparameter tuning in the number of linear layers and neurons per layer (increasing these to two hidden layers of size 64 and three linear layers improved results), the vocabulary size and sequence length (both of which were originally too high), and balancing the batch size and learning rate (our best results had a batch size of 32 and learning rate .0005).

This architecture was tried with a normal LSTM and with a BiLSTM; we found that the BiLSTM performed better and its results are reported in the “LSTM” parts of the Results section.

4.4 SBERT and Linear Network

We used a pre-trained variation on BERT known as Sentence-BERT (SBERT), followed by a linear network. The linear network had 5 fully connected layers with a ReLU and dropout between them.

As there are multiple variations on SBERT, we were able to test some of these; we found that the largest one (“all-mpnet-base-v2”) worked best.

4.5 RoBERTa

Here, we used RoBERTa with the last four layers unfrozen; this was followed by a linear network that used batch normalization, took in the embedding of size 768, and halved the number of nodes at each layer to create a deep network.

4.6 GPT-3

We fine-tuned several GPT-3 engines (ada, babbage, curie, and davinci) using a portion of the training data due to cost constraints. We adjusted the training set size for each engine to ensure that cost was the same across all engines and found that the babbage model gave the most cost-effective performance.

The prompt was made by combining the post title and body, and the completion was the predicted label. We had our final output as the probability of generating a 0 or 1.

5 Results

To evaluate all the different binary classifiers we constructed, we assessed each model’s AUC score, accuracy score, and F1 score. Because most of our data had label 0, we were especially focused on the AUC and F1 scores. Given that the models output a probability for each text belonging to the positive class, we needed to select a threshold before classifying each text into a binary positive/negative classification before measuring the accuracy F1 score. We selected these thresholds by selecting the one that maximized these scores on the training set. The metrics for each model are reported below.

Model	AUC		Accuracy		F1	
	Training	Testing	Training	Testing	Training	Testing
Roberta	0.720	0.696	75.43%	74.02%	0.5191	0.4868
SBert	0.695	0.673	73.80%	63.18%	0.4958	0.4674
LGBM	0.853	0.636	81.86%	70.23%	0.6527	0.4009

LSTM	0.594	0.576	72.83%	73.02%	0.4408	0.4308
CNN	0.865	0.514	82.53%	63.38%	0.6721	0.3007
GPT-3	—	0.640*	—	71.90%*	—	0.4732*
Random Baseline	0.5		73.0%		0.422	

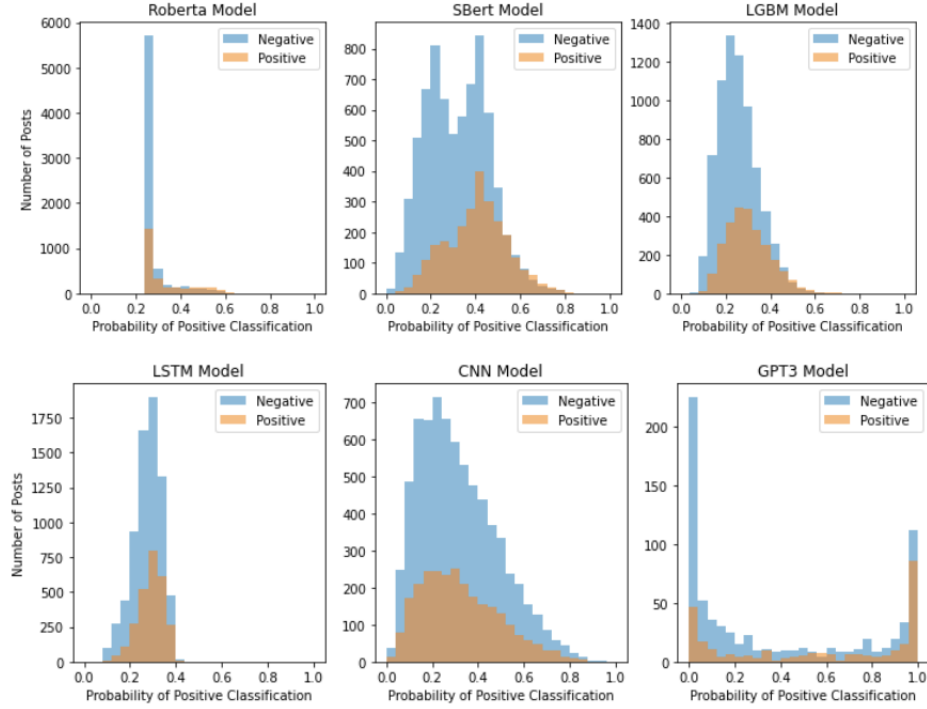
*Due to computational resources, we were only able to calculate the test set accuracy for the GPT-3 model on a random sample of 1000 texts

We find that the models generally had a difficult time distinguishing the two classes apart. The fine-tuned roberta model performed best with the highest scores across all three metrics in the testing set. In terms of AUC, the Sbert model achieved the second highest score followed by the GPT-3 model, LGBM model, LSTM model, and then the CNN model which achieved a score of 0.514 which is only slightly above random chance. The F1 scores follow a similar pattern, with the difference being that the LSTM model overtakes the LGBM model. In terms of accuracy, the LSTM model performs the second best, followed by the LGBM model, the CNN model, and then the Sbert model. We note that the baseline accuracy, which is to always predict the negative class, is 73.0%, the only model to get a score better than this is the Roberta model. The Roberta model only attains a single percentage point above this baseline. The baseline F1 score, on the other hand is 0.422, and the GPT-3, Roberta, Sbert, and LSTM models were able to achieve F1 score greater than this baseline.

Comparing the results across the testing and training sets, we see that the SBert and CNN models are overfitting significantly. The CNN model especially overfits as we observe a drop from 0.865 to 0.514 in the AUC.

5.1 Error Analysis

The histograms below show the distribution of positive classification probabilities for the positive and negative labels in the dataset. We note that the best performing models (Roberta, SBert) include a region where they’re able to accurately identify some examples as negative well within the low-probability section of the histogram. As the probability for a positive classification increases, we see that models begin to have a hard time distinguishing positive cases from negative cases. We also note that the models are generally very unconfident. The Roberta, LGBM, and LSTM models for example, places most of its probability assignments around 0.2-0.3, with a tiny minority of cases being assigned a higher probability. We observe the opposite effect with the GPT-3 model which confidently assigns most labels as being very near 0 or 1.



The chart below attempts to explore different variables that could account for differences in the model’s AUC. We examine four different variables. We first examine to see if models with a text length greater than 2000 characters obtain a worse AUC given that for the pretrained models, texts greater than this length are effectively truncated to fit within each model’s token limit. We also examine to see if edited posts obtain a higher AUC under the assumption that edited posts might contain information from “the future” meaning that a post might contain information about the label assigned to it. Finally, we examine to see if the amount of comments and the number of upvotes on a post affect the AUC under the assumption that posts with higher values for these variables have been subjected to scrutiny from larger amounts of people which might decrease the “randomness” in the ground truth label. We note that the number of posts in each category are large enough as to be statistically significant ($n \geq 3000$ for each AUC calculated). We selected the cutoff for the comment amount and post score such that each bin contains the same number of posts.

Model	Text Length		Edited		# of comments		Post Score	
	≤ 2000	>2000	False	True	≤ 23	>23	≤ 11	>11
Roberta	0.688	0.709	0.697	0.677	0.684	0.687	0.693	0.705

SBert	0.668	0.681	0.683	0.646	0.661	0.666	0.663	0.689
LGBM	0.637	0.633	0.636	0.607	0.622	0.629	0.627	0.651
LSTM	0.573	0.573	0.584	0.552	0.565	0.578	0.563	0.593
CNN	0.520	0.501	0.522	0.497	0.515	0.511	0.516	0.512

For the text length, we find the counterintuitive result that longer texts obtain a higher AUC. This is especially counterintuitive for the Roberta and SBert models given that longer texts are effectively truncated. A possible explanation for this may be that longer texts effectively contain “more information” that the model can use for its decision. Additionally, we note a slight negative correlation (-0.082,- 0.103) between text length and the probabilities for a positive classification for the Roberta and Sbert model. This indicates that these models might be using the text length as a signal indicating a negative classification.

A similar counterintuitive result was found for edited texts where we see that the classification AUC for edited texts is lower than non-edited texts. We note that there’s a slight correlation (0.09) between a label being positive and the text being edited. Thus it’s possible that the difference comes from the fact that the models obtain a worse AUC when the proportion of positive labels is higher in the test set.

Comparing the AUC for posts with different scores and comments, we find that the models perform better on posts with a higher score, but that there’s no difference in terms of AUC for posts based on the number of comments. Given that both variables serve as a proxy for the scrutiny for the post label, we expected both variables to reflect a similar relationship to the AUC. We have no good explanations for this discrepancy.

6 Discussion

6.1 Findings & Limitations

Overall, we find that the task of predicting the YTA/NTA label for the posts is difficult. In line with what the previous study[6] predicting norms from Goofus Gallant comics found, we saw that models relying on pre-trained language models (RoBERTa, SBERT) performed well. The method that performed best in their study (fine tuning a BERT-based model) also performed well in our case. In our study, we also found that a gradient-boosted relying on a bag of words representation of the text outperformed an LSTM and CNN model on our main metric of evaluation (AUC) may be either due to the fact that we did not have enough data to train these models, or that the architectures we used were not optimal.

Comparing the performances of these models, we note that we find meaningful differences in their performance as each model becomes more capable and thus suited to the task. It’s likely that the LGBM model is able to perform above-baseline in the AUC by being able to pick up shallow linguistic signals and patterns. For instance, it may be able to pick up on very subtle patterns, like the fact that writers that are in the wrong may have more incentive to edit. The SBert and the Roberta model improve on this by being able to pick up on more complicated patterns. As a whole however, we have no guesses as to what these patterns look like as we did not perform any deeper analysis into interpreting the outputs of these transformer models.

We also note the subpar-performance of the GPT-3 based model, which may be due to the fact that we only provided it with a fraction of the training data provided to the rest of the models due to cost constraints. In any case, we have shown that GPT-3’s 2500-shot performance on classifying these posts is not competitive with other SOTA models. There is still a lot to explore however in determining whether or not these types of generative models can perform this type of task. We have yet to examine GPT-3’s performance if we fine tune it with more data, or further optimize it by utilizing chain-of-logic prompting which has been shown to improve its performance on complicated tasks.

6.2 Ethical Considerations

It’s important to emphasize that the data we use in training and evaluating our models is not perfect. While it captures the aggregate moral decisions of the reddit’s r/amatheasshole community, there are no guarantees that this community makes good moral decisions. In addition to this, we did not perform any in depth analysis to ensure that the decisions made by the community are consistent and free of bias. Thus, even if our model did perform excellently, it only means that it can capture the decision making process of the community, not that it can serve as a good model for making any moral decisions.

Furthermore, even if our model or a future model that performs the same task performs excellently on the prediction task for a fixed dataset, norms can shift and vary over time. Some ideas once viewed as acceptable in the past can be seen as reprehensible today and some ideas viewed as reprehensible are now commonly held. This means that any model trained to ”model morality” must be able to update to reflect the ability of culture to evolve.

6.3 Future Research Directions

Overall, we find that the task of identifying the label of these morality stories is difficult yet feasible to some degree. We believe due to the challenge it presents, this dataset can serve as a useful benchmark for evaluating different language models as it can essentially represent a model’s capability to replicate the collective decision made by a given community.

Additionally, given both resource and time constraints, it’s unlikely that we were able to optimize and unlock the full potential of the models we examined. Future work can extend our study by fine tuning on a larger pretrained model for example, or running GPT-3 with more examples and better prompting.

There are also several questions left to be answered about how any of our models, including our transformer based models, make their decisions. A study looking into things like looking at what the attention heads in these models attend to can help interpret how they determine morality. Such work would be especially important if these kinds of models would be used as components of other systems that need some way to measure morality.

7 Conclusion

In this study, we utilized posts in the “Am I The Asshole” Reddit forum as training data to train different deep learning models to make moral judgments. We compared and analyzed the efficacy of five deep learning models including fine-tuned Roberta, SBert, GPT-3, convolutional neural network, and Bi-LSTM. Based on our results, a fine-tuned Roberta model achieved the best AUC of 0.70. Our study demonstrates the potential of using posts in social media communities like Reddit AITA forum as a potential dataset for benchmarking agents that aim to achieve value alignment with NLP techniques.

8 References

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